IT Fundamentals: Command Cheatsheet

Two reference tables: basic Windows network commands, and user, group and permission commands (Linux equivalents marked).

Network Commands (Windows Basics)

The core troubleshooting order is: check the IP configuration, ping the local gateway, ping an outside IP, then test DNS by name.

Command	Use
`ipconfig`	Show IP address, subnet mask, default gateway
`ipconfig /all`	Full detail: MAC address, DHCP server, DNS servers, lease times
`ipconfig /release`	Drop the current DHCP lease
`ipconfig /renew`	Request a new DHCP lease
`ipconfig /flushdns`	Clear the DNS resolver cache
`ipconfig /displaydns`	View cached DNS entries
`ping <host>`	Test reachability and latency (ICMP)
`ping 127.0.0.1`	Test the local TCP/IP stack (loopback)
`ping -t <host>`	Continuous ping until stopped with Ctrl+C
`tracert <host>`	Show each router hop to a destination; find where traffic stops
`pathping <host>`	Combines ping and tracert with packet loss per hop
`nslookup <name>`	Query DNS; resolve a name to an IP
`nslookup <name> 8.8.8.8`	Query a specific DNS server to test your own
`netstat -an`	List open ports and connections
`netstat -b`	Show which program owns each connection (admin)
`arp -a`	Show IP-to-MAC address table
`route print`	Show the routing table
`getmac`	Show the NIC's MAC address
`hostname`	Show the computer's name
`nbtstat -n`	Show NetBIOS names
`netsh interface ip show config`	Show IP settings per adapter
`netsh interface ip set address "Ethernet" static <ip> <mask> <gw>`	Set a static IP

Command	Use
`netsh interface ip set address "Ethernet" dhcp`	Switch an adapter back to DHCP
`netsh interface ip set dns "Ethernet" static <ip>`	Set the DNS server
`netsh wlan show profiles`	List saved Wi-Fi networks
`netsh winsock reset`	Reset the network stack (restart afterwards)
`Test-NetConnection <host> -Port 443` (PowerShell)	Test a specific TCP port

Users, Groups and Permissions

These cover managing user accounts, groups, and file or folder permissions. Run in an elevated (Administrator) prompt where noted.

Users, groups and permissions

Command	Use
`whoami`	Show the current user
`net user`	List local user accounts
`net user <name> <password> /add`	Create a user
`net user <name> /delete`	Delete a user
`net user <name> *`	Change a password (prompts securely)
`net localgroup administrators <name> /add`	Make a user an administrator
`lusrmgr.msc`	Local Users and Groups (Pro editions)
`netplwiz`	User accounts and auto-login settings
`icacls <path>`	View NTFS permissions
`icacls <path> /grant <user>:(OI)(CI)F`	Give full control, inherited by subfolders and files
`icacls <path> /remove <user>`	Remove a user's permissions
`takeown /f <path> /r`	Take ownership of a file or folder
`attrib +r +h <file>`	Set read-only and hidden attributes
`attrib -r -h <file>`	Clear those attributes
`secpol.msc`	Local Security Policy: password and lockout rules
`gpedit.msc`	Local Group Policy Editor (Pro editions)
`gpupdate /force`	Apply Group Policy changes immediately
`gpresult /r`	Show which policies apply to this user and PC
`id` / `groups` (Linux)	Show user ID and group membership
`chmod 755 <file>` (Linux)	Set permissions with numbers
`chown user:group <file>` (Linux)	Change owner and group
`sudo <command>` (Linux)	Run as administrator

Linux permission numbers: read = 4, write = 2, execute = 1, added per owner/group/others. So 755 = owner full, others read and execute; 644 = owner read/write, others read only.